



1. Description

1.1. Project

Project Name	P2
Board Name	P2
Generated with:	STM32CubeMX 6.16.1
Date	01/23/2026

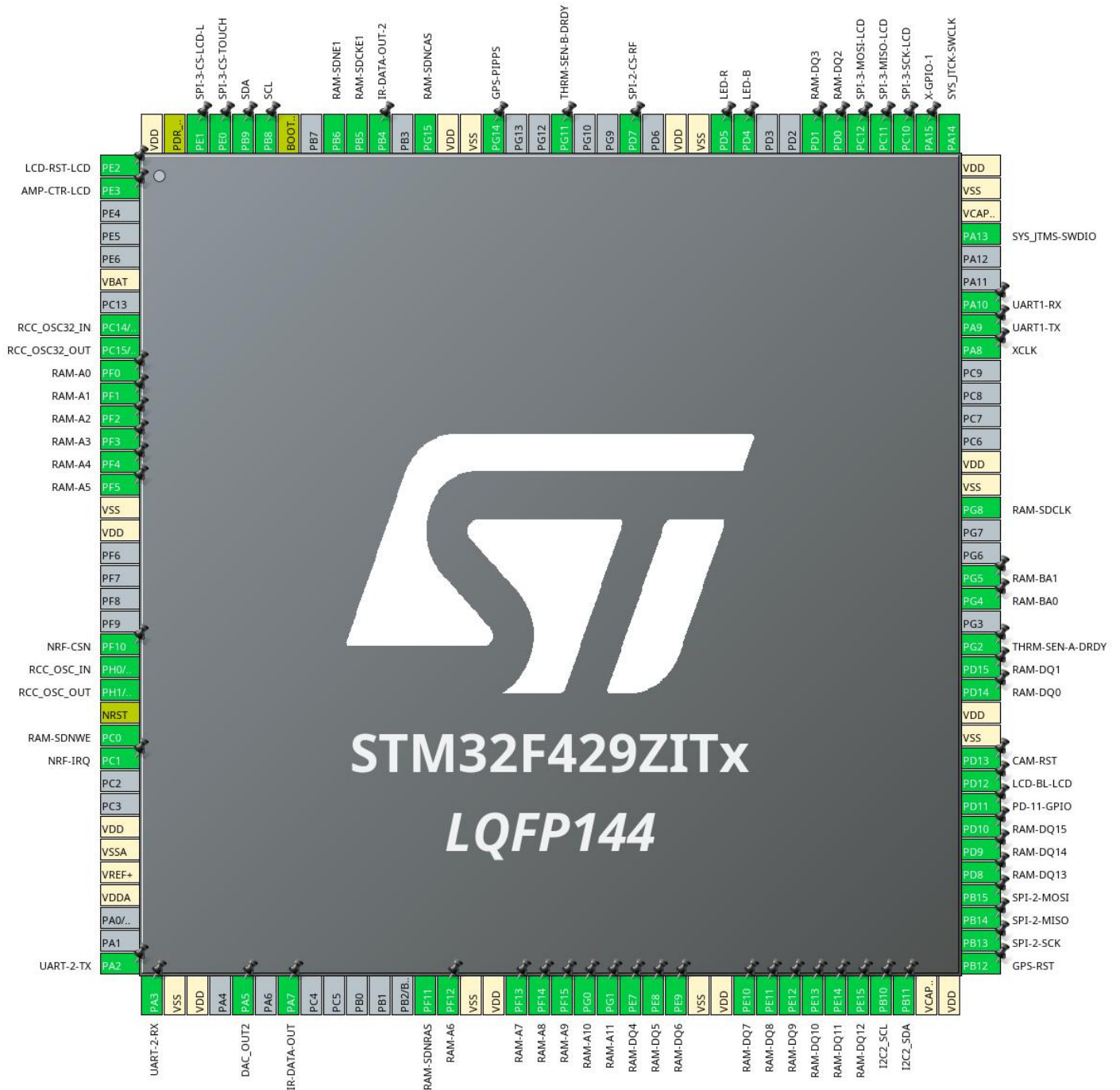
1.2. MCU

MCU Series	STM32F4
MCU Line	STM32F429/439
MCU name	STM32F429ZITx
MCU Package	LQFP144
MCU Pin number	144

1.3. Core(s) information

Core(s)	ARM Cortex-M4
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2. Pinout Configuration



3. Pins Configuration

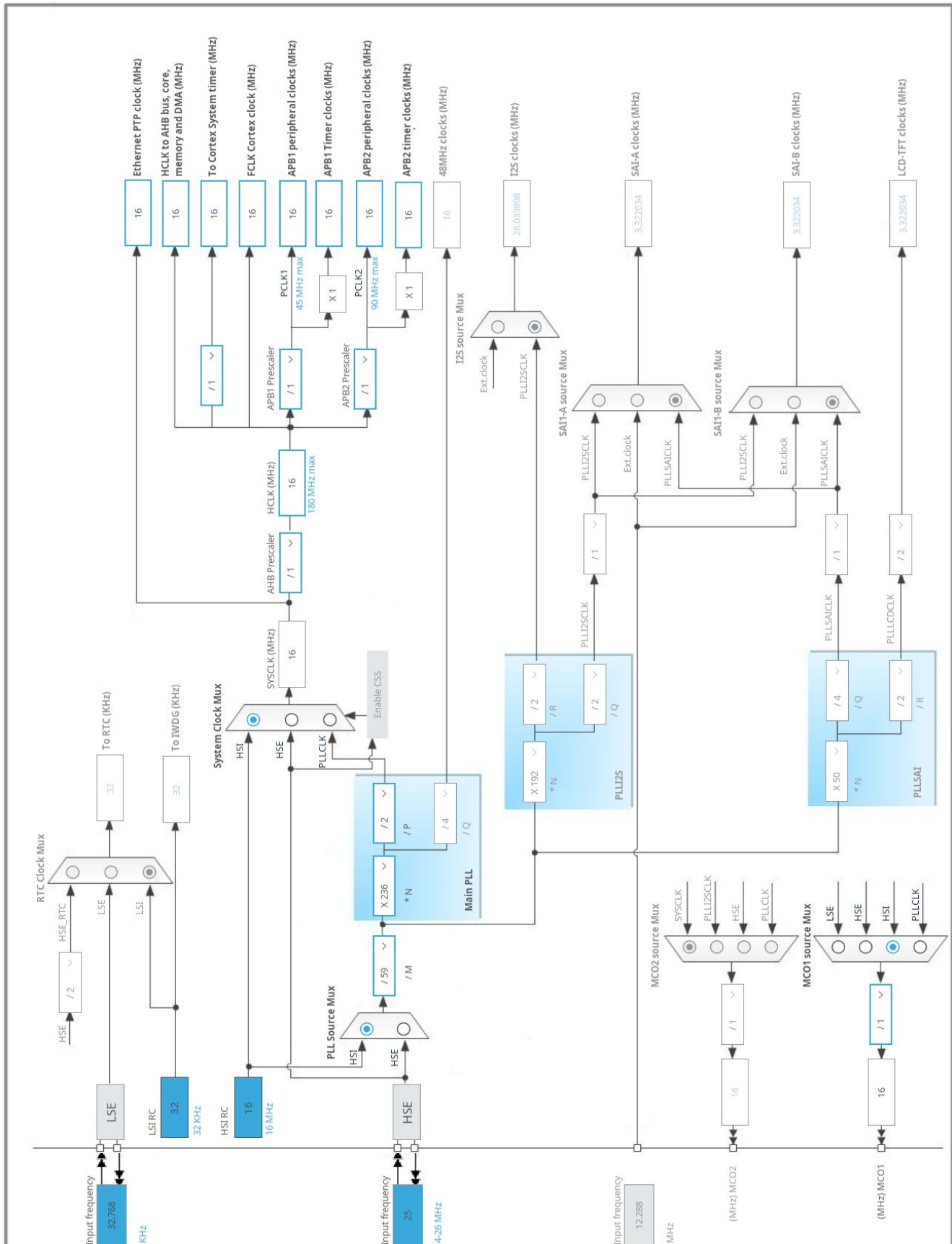
Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
1	PE2 *	I/O	GPIO_Output	LCD-RST-LCD
2	PE3 *	I/O	GPIO_Output	AMP-CTR-LCD
6	VBAT	Power		
8	PC14/OSC32_IN	I/O	RCC_OSC32_IN	
9	PC15/OSC32_OUT	I/O	RCC_OSC32_OUT	
10	PF0	I/O	FMC_A0	RAM-A0
11	PF1	I/O	FMC_A1	RAM-A1
12	PF2	I/O	FMC_A2	RAM-A2
13	PF3	I/O	FMC_A3	RAM-A3
14	PF4	I/O	FMC_A4	RAM-A4
15	PF5	I/O	FMC_A5	RAM-A5
16	VSS	Power		
17	VDD	Power		
22	PF10 *	I/O	GPIO_Output	NRF-CSN
23	PH0/OSC_IN	I/O	RCC_OSC_IN	
24	PH1/OSC_OUT	I/O	RCC_OSC_OUT	
25	NRST	Reset		
26	PC0	I/O	FMC_SDNWE	RAM-SDNWE
27	PC1	I/O	GPIO_EXTI1	NRF-IRQ
30	VDD	Power		
31	VSSA	Power		
32	VREF+	Power		
33	VDDA	Power		
36	PA2	I/O	USART2_TX	UART-2-TX
37	PA3	I/O	USART2_RX	UART-2-RX
38	VSS	Power		
39	VDD	Power		
41	PA5	I/O	DAC_OUT2	
43	PA7	I/O	TIM3_CH2	IR-DATA-OUT
49	PF11	I/O	FMC_SDNRAS	RAM-SDNRAS
50	PF12	I/O	FMC_A6	RAM-A6
51	VSS	Power		
52	VDD	Power		
53	PF13	I/O	FMC_A7	RAM-A7
54	PF14	I/O	FMC_A8	RAM-A8
55	PF15	I/O	FMC_A9	RAM-A9

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
56	PG0	I/O	FMC_A10	RAM-A10
57	PG1	I/O	FMC_A11	RAM-A11
58	PE7	I/O	FMC_D4	RAM-DQ4
59	PE8	I/O	FMC_D5	RAM-DQ5
60	PE9	I/O	FMC_D6	RAM-DQ6
61	VSS	Power		
62	VDD	Power		
63	PE10	I/O	FMC_D7	RAM-DQ7
64	PE11	I/O	FMC_D8	RAM-DQ8
65	PE12	I/O	FMC_D9	RAM-DQ9
66	PE13	I/O	FMC_D10	RAM-DQ10
67	PE14	I/O	FMC_D11	RAM-DQ11
68	PE15	I/O	FMC_D12	RAM-DQ12
69	PB10	I/O	I2C2_SCL	
70	PB11	I/O	I2C2_SDA	
71	VCAP_1	Power		
72	VDD	Power		
73	PB12 *	I/O	GPIO_Output	GPS-RST
74	PB13	I/O	SPI2_SCK	SPI-2-SCK
75	PB14	I/O	SPI2_MISO	SPI-2-MISO
76	PB15	I/O	SPI2_MOSI	SPI-2-MOSI
77	PD8	I/O	FMC_D13	RAM-DQ13
78	PD9	I/O	FMC_D14	RAM-DQ14
79	PD10	I/O	FMC_D15	RAM-DQ15
80	PD11 *	I/O	GPIO_Output	PD-11-GPIO
81	PD12 *	I/O	GPIO_Output	LCD-BL-LCD
82	PD13 *	I/O	GPIO_Output	CAM-RST
83	VSS	Power		
84	VDD	Power		
85	PD14	I/O	FMC_D0	RAM-DQ0
86	PD15	I/O	FMC_D1	RAM-DQ1
87	PG2	I/O	GPIO_EXTI2	THRM-SEN-A-DRDY
89	PG4	I/O	FMC_BA0	RAM-BA0
90	PG5	I/O	FMC_BA1	RAM-BA1
93	PG8	I/O	FMC_SDCLK	RAM-SDCLK
94	VSS	Power		
95	VDD	Power		
100	PA8	I/O	RCC_MCO_1	XCLK
101	PA9	I/O	USART1_TX	UART1-TX

Pin Number LQFP144	Pin Name (function after reset)	Pin Type	Alternate Function(s)	Label
102	PA10	I/O	USART1_RX	UART1-RX
105	PA13	I/O	SYS_JTMS-SWDIO	
106	VCAP_2	Power		
107	VSS	Power		
108	VDD	Power		
109	PA14	I/O	SYS_JTCK-SWCLK	
110	PA15 *	I/O	GPIO_Output	X-GPIO-1
111	PC10	I/O	SPI3_SCK	SPI-3-SCK-LCD
112	PC11	I/O	SPI3_MISO	SPI-3-MISO-LCD
113	PC12	I/O	SPI3_MOSI	SPI-3-MOSI-LCD
114	PD0	I/O	FMC_D2	RAM-DQ2
115	PD1	I/O	FMC_D3	RAM-DQ3
118	PD4 *	I/O	GPIO_Output	LED-B
119	PD5 *	I/O	GPIO_Output	LED-R
120	VSS	Power		
121	VDD	Power		
123	PD7 *	I/O	GPIO_Output	SPI-2-CS-RF
126	PG11	I/O	GPIO_EXTI11	THRM-SEN-B-DRDY
129	PG14	I/O	GPIO_EXTI14	GPS-PIPPS
130	VSS	Power		
131	VDD	Power		
132	PG15	I/O	FMC_SDNCAS	RAM-SDNCAS
134	PB4	I/O	TIM3_CH1	IR-DATA-OUT-2
135	PB5	I/O	FMC_SDCKE1	RAM-SDCKE1
136	PB6	I/O	FMC_SDNE1	RAM-SDNE1
138	BOOT0	Boot		
139	PB8	I/O	I2C1_SCL	SCL
140	PB9	I/O	I2C1_SDA	SDA
141	PE0 *	I/O	GPIO_Output	SPI-3-CS-TOUCH
142	PE1 *	I/O	GPIO_Output	SPI-3-CS-LCD-L
143	PDR_ON	Reset		
144	VDD	Power		

* The pin is affected with an I/O function

4. Clock Tree Configuration



1. Software Project

1.1. Project Settings

Name	Value
Project Name	P2
Project Folder	/home/musman/Desktop/C/ROBOT_FW
Toolchain / IDE	MDK-ARM V5
Firmware Package Name and Version	STM32Cube FW_F4 V1.9.0
Application Structure	Basic
Generate Under Root	No
Do not generate the main()	No
Minimum Heap Size	0x200
Minimum Stack Size	0x400

1.2. Code Generation Settings

Name	Value
STM32Cube MCU packages and embedded software	Copy all used libraries into the project folder
Generate peripheral initialization as a pair of '.c/.h' files	No
Backup previously generated files when re-generating	Yes
Keep User Code when re-generating	Yes
Delete previously generated files when not re-generated	Yes
Set all free pins as analog (to optimize the power consumption)	No
Enable Full Assert	No

1.3. Advanced Settings - Generated Function Calls

Rank	Function Name	Peripheral Instance Name
1	SystemClock_Config	RCC
2	MX_GPIO_Init	GPIO
3	MX_DMA_Init	DMA
4	MX_DAC_Init	DAC
5	MX_FMC_Init	FMC
6	MX_I2C1_Init	I2C1
7	MX_I2C2_Init	I2C2
8	MX_SPI2_Init	SPI2
9	MX_SPI3_Init	SPI3
10	MX_TIM3_Init	TIM3
11	MX_TIM6_Init	TIM6

Rank	Function Name	Peripheral Instance Name
12	MX_USART1_UART_Init	USART1
13	MX_USART2_UART_Init	USART2

2. Peripherals and Middlewares Configuration

2.1. DAC

mode: OUT2 Configuration

2.1.1. Parameter Settings:

DAC Out2 Settings:

Output Buffer	Enable
Trigger	Timer 6 Trigger Out event *
Wave generation mode	Triangle wave generation *
Maximum Triangle Amplitude	4095 *

2.2. FMC

SDRAM 1

Clock and chip enable: SDCKE1+SDNE1

Internal bank number: 4 banks

Address: 12 bits

Data: 16 bits

2.2.1. SDRAM 1:

SDRAM control:

Bank	SDRAM bank 2
Column bit number	8 bits
Row bit number	11 bits
CAS latency	1 memory clock cycle
Write protection	Disabled
SDRAM common clock	Disabled
SDRAM common burst read	Disabled
SDRAM common read pipe delay	0 HCLK clock cycle

SDRAM timing in memory clock cycles:

Load mode register to active delay	16
Exit self-refresh delay	16
Self refresh time	16
SDRAM common row cycle delay	16
Write recovery time	16
SDRAM common row precharge delay	16
Row to column delay	16

2.3. I2C1

I2C: I2C

2.3.1. Parameter Settings:

Master Features:

I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

2.4. I2C2

I2C: I2C

2.4.1. Parameter Settings:

Master Features:

I2C Speed Mode	Standard Mode
I2C Clock Speed (Hz)	100000

Slave Features:

Clock No Stretch Mode	Disabled
Primary Address Length selection	7-bit
Dual Address Acknowledged	Disabled
Primary slave address	0
General Call address detection	Disabled

2.5. RCC

High Speed Clock (HSE): Crystal/Ceramic Resonator

Low Speed Clock (LSE) : Crystal/Ceramic Resonator

mode: Master Clock Output 1

2.5.1. Parameter Settings:

System Parameters:

VDD voltage (V)	3.3
Instruction Cache	Enabled
Prefetch Buffer	Enabled
Data Cache	Enabled
Flash Latency(WS)	0 WS (1 CPU cycle)

RCC Parameters:

HSI Calibration Value	0 *
TIM Prescaler Selection	Disabled

Power Parameters:

Power Regulator Voltage Scale	Power Regulator Voltage Scale 3
Power Over Drive	Disabled

2.6. SPI2

Mode: Full-Duplex Master

2.6.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
Data Size	8 Bits
First Bit	MSB First

Clock Parameters:

Prescaler (for Baud Rate)	2
Baud Rate	8.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge

Advanced Parameters:

CRC Calculation	Disabled
NSS Signal Type	Software

2.7. SPI3

Mode: Full-Duplex Master

2.7.1. Parameter Settings:

Basic Parameters:

Frame Format	Motorola
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Data Size	8 Bits
First Bit	MSB First
Clock Parameters:	
Prescaler (for Baud Rate)	2
Baud Rate	8.0 MBits/s *
Clock Polarity (CPOL)	Low
Clock Phase (CPHA)	1 Edge
Advanced Parameters:	
CRC Calculation	Disabled
NSS Signal Type	Software

2.8. SYS

Debug: Serial Wire Debug (SWD)

2.9. TIM3

Slave Mode: Reset Mode

Trigger Source: TI1FP1

Clock Source : Internal Clock

Channel1: Input Capture direct mode

Channel2: Input Capture direct mode

2.9.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	16000 *
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	19 *
Internal Clock Division (CKD)	No Division
Slave Mode Controller	Reset Mode

Trigger Output (TRGO) Parameters:

Master/Slave Mode	Disable (no sync between this TIM (Master) and its Slaves)
Trigger Event Selection	Update Event *

Input Capture Channel 1:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

Input Capture Channel 2:

Polarity Selection	Rising Edge
IC Selection	Direct
Prescaler Division Ratio	No division
Input Filter (4 bits value)	0

2.10. TIM6

mode: Activated

2.10.1. Parameter Settings:

Counter Settings:

Prescaler (PSC - 16 bits value)	0
Counter Mode	Up
Counter Period (AutoReload Register - 16 bits value)	5 *

Trigger Output (TRGO) Parameters:

Trigger Event Selection	Update Event *
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2.11. USART1

Mode: Asynchronous

2.11.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

2.12. USART2

Mode: Asynchronous

2.12.1. Parameter Settings:

Basic Parameters:

Baud Rate	115200
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Word Length	8 Bits (including Parity)
Parity	None
Stop Bits	1

Advanced Parameters:

Data Direction	Receive and Transmit
Over Sampling	16 Samples

*** User modified value**

3. System Configuration

3.1. GPIO configuration

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
DAC	PA5	DAC_OUT2	Analog mode	No pull-up and no pull-down	n/a	
FMC	PF0	FMC_A0	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A0
	PF1	FMC_A1	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A1
	PF2	FMC_A2	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A2
	PF3	FMC_A3	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A3
	PF4	FMC_A4	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A4
	PF5	FMC_A5	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A5
	PC0	FMC_SDNWE	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDNWE
	PF11	FMC_SDNRAS	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDNRAS
	PF12	FMC_A6	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A6
	PF13	FMC_A7	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A7
	PF14	FMC_A8	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A8
	PF15	FMC_A9	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A9
	PG0	FMC_A10	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A10
	PG1	FMC_A11	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-A11
	PE7	FMC_D4	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ4
	PE8	FMC_D5	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ5
	PE9	FMC_D6	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ6
	PE10	FMC_D7	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ7
	PE11	FMC_D8	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ8
	PE12	FMC_D9	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ9
	PE13	FMC_D10	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ10
	PE14	FMC_D11	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ11
	PE15	FMC_D12	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ12
	PD8	FMC_D13	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ13
	PD9	FMC_D14	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ14
	PD10	FMC_D15	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ15
	PD14	FMC_D0	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ0
	PD15	FMC_D1	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ1
	PG4	FMC_BA0	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-BA0
	PG5	FMC_BA1	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-BA1
	PG8	FMC_SDCLK	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDCLK
	PD0	FMC_D2	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ2
	PD1	FMC_D3	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-DQ3
	PG15	FMC_SDNCAS	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDNCAS
	PB5	FMC_SDCKE1	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDCKE1

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
	PB6	FMC_SDNE1	Alternate Function Push Pull	No pull-up and no pull-down	High	RAM-SDNE1
I2C1	PB8	I2C1_SCL	Alternate Function Open Drain	Pull-up	Medium *	SCL
	PB9	I2C1_SDA	Alternate Function Open Drain	Pull-up	Medium *	SDA
I2C2	PB10	I2C2_SCL	Alternate Function Open Drain	No pull-up and no pull-down *	Fast *	
	PB11	I2C2_SDA	Alternate Function Open Drain	No pull-up and no pull-down *	Fast *	
RCC	PC14/OSC32_IN	RCC_OSC32_IN	n/a	n/a	n/a	
	PC15/OSC32_OUT	RCC_OSC32_OUT	n/a	n/a	n/a	
	PH0/OSC_IN	RCC_OSC_IN	n/a	n/a	n/a	
	PH1/OSC_OUT	RCC_OSC_OUT	n/a	n/a	n/a	
	PA8	RCC_MCO_1	Alternate Function Push Pull	No pull-up and no pull-down	Fast *	XCLK
SPI2	PB13	SPI2_SCK	Alternate Function Push Pull	No pull-up and no pull-down	High *	SPI-2-SCK
	PB14	SPI2_MISO	Alternate Function Push Pull	No pull-up and no pull-down	High *	SPI-2-MISO
	PB15	SPI2_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	High *	SPI-2-MOSI
SPI3	PC10	SPI3_SCK	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	SPI-3-SCK-LCD
	PC11	SPI3_MISO	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	SPI-3-MISO-LCD
	PC12	SPI3_MOSI	Alternate Function Push Pull	No pull-up and no pull-down	Medium *	SPI-3-MOSI-LCD
SYS	PA13	SYS_JTMS-SWDIO	n/a	n/a	n/a	
	PA14	SYS_JTCK-SWCLK	n/a	n/a	n/a	
TIM3	PA7	TIM3_CH2	Alternate Function Push Pull	No pull-up and no pull-down	Low	IR-DATA-OUT
	PB4	TIM3_CH1	Alternate Function Push Pull	No pull-up and no pull-down	Low	IR-DATA-OUT-2
USART1	PA9	USART1_TX	Alternate Function Push Pull	Pull-up	High *	UART1-TX
	PA10	USART1_RX	Alternate Function Push Pull	Pull-up	High *	UART1-RX
USART2	PA2	USART2_TX	Alternate Function Push Pull	Pull-up	Low	UART-2-TX
	PA3	USART2_RX	Alternate Function Push Pull	Pull-up	Low	UART-2-RX
GPIO	PE2	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LCD-RST-LCD
	PE3	GPIO_Output	Output Open Drain *	No pull-up and no pull-down	Low	AMP-CTR-LCD
	PF10	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	NRF-CSN
	PC1	GPIO_EXTI1	External Interrupt Mode with Falling	No pull-up and no pull-down	n/a	NRF-IRQ

IP	Pin	Signal	GPIO mode	GPIO pull/up pull down	Max Speed	User Label
			edge trigger detection			
	PB12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	GPS-RST
	PD11	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	PD-11-GPIO
	PD12	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LCD-BL-LCD
	PD13	GPIO_Output	Output Push Pull	Pull-down *	Low	CAM-RST
	PG2	GPIO_EXTI2	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	THRM-SEN-A-DRDY
	PA15	GPIO_Output	Output Push Pull	Pull-down *	Low	X-GPIO-1
	PD4	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED-B
	PD5	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	LED-R
	PD7	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI-2-CS-RF
	PG11	GPIO_EXTI11	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	THRM-SEN-B-DRDY
	PG14	GPIO_EXTI14	External Interrupt Mode with Rising edge trigger detection	No pull-up and no pull-down	n/a	GPS-PIPPS
	PE0	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI-3-CS-TOUCH
	PE1	GPIO_Output	Output Push Pull	No pull-up and no pull-down	Low	SPI-3-CS-LCD-L

3.2. DMA configuration

DMA request	Stream	Direction	Priority
SPI2_RX	DMA1_Stream3	Peripheral To Memory	Low
SPI2_TX	DMA1_Stream4	Memory To Peripheral	Low
I2C1_RX	DMA1_Stream0	Peripheral To Memory	Low
USART2_RX	DMA1_Stream5	Peripheral To Memory	Low

SPI2_RX: DMA1_Stream3 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

SPI2_TX: DMA1_Stream4 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

I2C1_RX: DMA1_Stream0 DMA request Settings:

Mode: Normal
 Use fifo: Disable
 Peripheral Increment: Disable
 Memory Increment: **Enable ***
 Peripheral Data Width: Byte
 Memory Data Width: Byte

USART2_RX: DMA1_Stream5 DMA request Settings:

Mode: Normal
 Use fifo: Disable

Peripheral Increment: Disable
Memory Increment: **Enable ***
Peripheral Data Width: Byte
Memory Data Width: Byte

3.3. NVIC configuration

3.3.1. NVIC

Interrupt Table	Enable	Preenmption Priority	SubPriority
Non maskable interrupt	true	0	0
Memory management fault	true	0	0
Pre-fetch fault, memory access fault	true	0	0
Undefined instruction or illegal state	true	0	0
Debug monitor	true	0	0
System tick timer	true	0	0
RCC global interrupt	true	0	0
EXTI line1 interrupt	true	0	0
EXTI line2 interrupt	true	0	0
DMA1 stream0 global interrupt	true	0	0
DMA1 stream3 global interrupt	true	0	0
DMA1 stream4 global interrupt	true	0	0
DMA1 stream5 global interrupt	true	0	0
TIM3 global interrupt	true	0	0
USART1 global interrupt	true	0	0
EXTI line[15:10] interrupts	true	0	0
PVD interrupt through EXTI line 16	unused		
Flash global interrupt	unused		
I2C1 event interrupt	unused		
I2C1 error interrupt	unused		
I2C2 event interrupt	unused		
I2C2 error interrupt	unused		
SPI2 global interrupt	unused		
USART2 global interrupt	unused		
FMC global interrupt	unused		
SPI3 global interrupt	unused		
TIM6 global interrupt, DAC1 and DAC2 underrun error interrupts	unused		

3.3.2. NVIC Code generation

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
Non maskable interrupt	false	false	false
Memory management fault	false	false	false
Pre-fetch fault, memory access fault	false	false	false
Undefined instruction or illegal state	false	false	false
Debug monitor	false	false	false

Enabled interrupt Table	Select for init sequence ordering	Generate IRQ handler	Call HAL handler
System tick timer	false	true	true
RCC global interrupt	false	true	true
EXTI line1 interrupt	false	true	true
EXTI line2 interrupt	false	true	true
DMA1 stream0 global interrupt	false	true	true
DMA1 stream3 global interrupt	false	true	true
DMA1 stream4 global interrupt	false	true	true
DMA1 stream5 global interrupt	false	true	true
TIM3 global interrupt	false	true	true
USART1 global interrupt	false	true	true
EXTI line[15:10] interrupts	false	true	true

*** User modified value**

4. System Views

4.1. Category view

4.1.1. Current

Middleware						
System Core	Analog	Timers	Connectivity	Multimedia	Security	Computing
DMA ✓	DAC ✓	TIM3 ✓	FMC ✓			
GPIO ✓		TIM6 ✓	I2C1 ✓			
NVIC ✓			I2C2 ✓			
RCC ✓			SPI2 ✓			
SYS ✓			SPI3 ✓			
			USART1 ✓			
			USART2 ✓			

5. Docs & Resources

Type	Link
BSDL files	https://www.st.com/resource/en/bsdl_model/stm32f427-437_429-439_bsdI.zip
IBIS models	https://www.st.com/resource/en/ibis_model/stm32f427-437_429-439_ibis.zip
System View Description	https://www.st.com/resource/en/svd/stm32f4-svd.zip
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_embedded_software_solutions.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_eval_tools_portfolio.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32_stm8_functional-safety-packages.pdf
Presentations	https://www.st.com/resource/en/product_presentation/stm32-stm8_software_development_tools.pdf
Presentations	https://www.st.com/resource/en/product_presentation/microcontrollers-stm32-family-overview.pdf
Brochures	https://www.st.com/resource/en/brochure/products-and-solutions-for-plcs-and-smart-i-os.pdf
Brochures	https://www.st.com/resource/en/brochure/expansion-boards-for-intelligent-power-switches.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32nucleo.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32trust.pdf
Flyers	https://www.st.com/resource/en/flyer/flstm32gui.pdf
Product Certifications	https://www.st.com/resource/en/certification_document/stm32_authentication_can.pdf
Security Bulletin	https://www.st.com/resource/en/technical_note/tn1489-security-bulletin-tn1489stpsirt-physical-attacks-on-stm32-and-stm32cube-firmware-stmicroelectronics.pdf
Application Notes	https://www.st.com/resource/en/application_note/an1709-emc-design-

guide-for-stm8-stm32-and-legacy-mcus-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an2945-stm8s-and-stm32-mcus-a-consistent-832bit-product-line-for-painless-migration-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3070-managing-the-driver-enable-signal-for-rs485-and-iolink-communications-with-the-stm32s-usart-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3126-audio-and-waveform-generation-using-the-dac-in-stm32-products-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3155-usart-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3156-usb-dfu-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3364-migration-and-compatibility-guidelines-for-stm32-microcontroller-applications-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3997-audio-playback-and-recording-using-the-stm32f4discovery-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an3998-pdm-audio-software-decoding-on-stm32-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4031-using-the-stm32f2-stm32f4-and-stm32f7-series-dma-controller-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4073-how-to-improve-adc-accuracy-when-using-stm32f2xx-and-stm32f4xx-microcontrollers-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4221-i2c-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4286-spi-protocol-used-in-the-stm32-bootloader-stmicroelectronics.pdf

Application Notes https://www.st.com/resource/en/application_note/an4488-getting-started-with-stm32f4xxx-mcu-hardware-development-stmicroelectronics.pdf

- Application Notes https://www.st.com/resource/en/application_note/an4547-migrating-from-stm32f407xx417xx-to-stm32f427xx429xx437xx439xx-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4640-peripherals-interconnections-on-stm32f4057xx-stm32f4157xx-stm32f42xxx-stm32f43xxx-stm32f446xx-and-stm32f469479xx-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4655-virtually-increasing-the-number-of-serial-communication-peripherals-in-stm32-applications-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4658-migration-of-applications-from-stm32f429439-lines-to-stm32f446-line-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4660-migration-of-microcontroller-applications-from-stm32f42xxxf43xxx-devices-to-stm32f7-series-devices-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4675-migration-of-microcontroller-applications-from-stm32f42xxxstm32f43xxx-to-stm32f469xxstm32f479xx-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4739-stm32cube-firmware-examples-for-stm32f4-series-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4750-handling-of-soft-errors-in-stm32-applications-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4776-generalpurpose-timer-cookbook-for-stm32-microcontrollers-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4803-highspeed-si-simulations-using-ibis-and-boardlevel-simulations-using-hyperlynx-si-on-stm32-mcus-and-mpus-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4850-stm32-mcus-spreadspectrum-clock-generation-principles-properties-and-implementation-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4989-stm32-microcontroller-debug-toolbox-stmicroelectronics.pdf
- Application Notes https://www.st.com/resource/en/application_note/an4995-using-an-electromyogram-technique-to-detect-muscle-activity-

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for related Tools [lcd-glass-driver-firmware-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an2656-stm32f10xxx-for-related-Tools-lcd-glass-driver-firmware-stmicroelectronics.pdf)
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for related Tools [interfacing-with-the-highdensity-stm32f10xxx-fsmc-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an2790-tft-lcd-interfacing-with-the-highdensity-stm32f10xxx-fsmc-stmicroelectronics.pdf)
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for related Tools [modes-and-their-applications-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3116-stm32s-adc-modes-and-their-applications-stmicroelectronics.pdf)
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for related Tools [receivers-for-infrared-remote-control-protocols-using-stm32f10xxx-](https://www.st.com/resource/en/application_note/an3174-implementing-receivers-for-infrared-remote-control-protocols-using-stm32f10xxx-microcontrollers-stmicroelectronics.pdf)
& Software [microcontrollers-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3174-implementing-receivers-for-infrared-remote-control-protocols-using-stm32f10xxx-microcontrollers-stmicroelectronics.pdf)

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for related Tools [direct-drive-using-the-stm32f10xx-fsmc-peripheral-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3241-qvga-tftlcd-direct-drive-using-the-stm32f10xx-fsmc-peripheral-stmicroelectronics.pdf)
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for related Tools [obtaining-iec-60335-class-b-certification-for-any-stm32-application-](https://www.st.com/resource/en/application_note/an3307-guidelines-for-obtaining-iec-60335-class-b-certification-for-any-stm32-application-stmicroelectronics.pdf)
& Software [stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3307-guidelines-for-obtaining-iec-60335-class-b-certification-for-any-stm32-application-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an3965-](https://www.st.com/resource/en/application_note/an3965-stm32f40xstm32f41x-inapplication-programming-using-the-usart-stmicroelectronics.pdf)
for related Tools [stm32f40xstm32f41x-inapplication-programming-using-the-usart-](https://www.st.com/resource/en/application_note/an3965-stm32f40xstm32f41x-inapplication-programming-using-the-usart-stmicroelectronics.pdf)
& Software [stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3965-stm32f40xstm32f41x-inapplication-programming-using-the-usart-stmicroelectronics.pdf)

Application Notes [https://www.st.com/resource/en/application_note/an3969-eeeprom-](https://www.st.com/resource/en/application_note/an3969-eeeprom-emulation-in-stm32f40xstm32f41x-microcontrollers-stmicroelectronics.pdf)
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Application Notes [https://www.st.com/resource/en/application_note/an3990-upgrading-](https://www.st.com/resource/en/application_note/an3990-upgrading-stm32f4discovery-board-firmware-using-a-usb-key-stmicroelectronics.pdf)
for related Tools [stm32f4discovery-board-firmware-using-a-usb-key-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an3990-upgrading-stm32f4discovery-board-firmware-using-a-usb-key-stmicroelectronics.pdf)
& Software

Application Notes [https://www.st.com/resource/en/application_note/an4044-floating-point-](https://www.st.com/resource/en/application_note/an4044-floating-point-unit-demonstration-on-stm32-microcontrollers-stmicroelectronics.pdf)
for related Tools [unit-demonstration-on-stm32-microcontrollers-stmicroelectronics.pdf](https://www.st.com/resource/en/application_note/an4044-floating-point-unit-demonstration-on-stm32-microcontrollers-stmicroelectronics.pdf)
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Application Notes https://www.st.com/resource/en/application_note/an4841-digital-signal-

for related Tools & Software	processing-for-stm32-microcontrollers-using-cmsis-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an4968-proprietary-code-read-out-protection-pcrop-on-stm32f72xxx-and-stm32f73xxx-microcontrollers-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5056-integration-guide-for-the-xcubesbsfu-stm32cube-expansion-package-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5360-getting-started-with-projects-based-on-the-stm32mp1-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5361-getting-started-with-projects-based-on-dualcore-stm32h7-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5394-getting-started-with-projects-based-on-the-stm32l5-series-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5418-how-to-build-a-simple-usbp-d-sink-application-with-stm32cubemx-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5426-migrating-graphics-middleware-projects-from-stm32cubemx-540-to-stm32cubemx-550-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5464-position-control-of-a-three-phase-permanent-magnet-motor-using-xcubemcsdk-or-xcubemcsdkful-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5564-getting-started-with-projects-based-on-dualcore-stm32wl-microcontrollers-in-stm32cubeide-stmicroelectronics.pdf
Application Notes for related Tools & Software	https://www.st.com/resource/en/application_note/an5698-adapting-the-xcubestl-functional-safety-package-for-stm32-iec-61508-compliant-to-other-safety-standards-stmicroelectronics.pdf
Application Notes for related Tools	https://www.st.com/resource/en/application_note/an5731-stm32cubemx-and-stm32cubeide-threadsafe-solution-stmicroelectronics.pdf

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User Manuals	https://www.st.com/resource/en/user_manual/um3461-stm32f4-series-ulcsaiec-607301603351-selftest-library-user-guide-stmicroelectronics.pdf